

CHANTYRYA, Russian Federation

WMO: 239290

Lat: 60.3216N

Lon: 64.2165E

Elev: 173

StdP: 14.60

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.5	-28.4	-38.2	0.6	-33.2	-32.9	0.9	-28.4	16.4	0.6	15.0	9.1	2.5	310	0.639

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.1	84.9	67.9	81.6	66.1	77.9	64.2	69.9	81.2	67.9	78.3	66.0	75.3	6.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.0	96.4	74.3	64.0	89.8	73.5	62.0	83.6	71.6	34.0	81.1	32.4	78.6	30.8	75.5	78.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
17.6	15.1	13.5	-40.4	90.3	7.0	2.2	-45.4	91.8	-49.5	93.1	-53.5	94.3	-58.5	95.9		
			-40.5	71.8	6.9	2.0	-45.5	73.3	-49.5	74.5	-53.4	75.6	-58.4	77.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	33.3	-0.7	3.9	21.0	35.6	48.6	60.5	64.6	58.6	48.3	34.8	16.8	6.3	(6)
(7)		DBStd	25.11	15.79	15.04	11.23	10.28	9.39	8.65	7.48	7.43	7.54	9.41	14.47	15.72	(7)
(8)		HDD50	7322	1572	1292	898	438	145	16	1	10	120	478	997	1355	(8)
(9)		HDD65	11733	2037	1712	1363	883	514	186	103	227	504	937	1447	1820	(9)
(10)		CDD50	1242	0	0	0	5	102	330	454	277	68	6	0	0	(10)
(11)		CDD65	178	0	0	0	0	6	50	91	29	2	0	0	0	(11)
(12)		CDH74	1696	0	0	0	0	96	453	881	254	12	0	0	0	(12)
(13)		CDH80	444	0	0	0	0	16	117	242	68	1	0	0	0	(13)
(14)	Wind	WSAvg	6.1	5.4	5.4	6.8	6.8	7.1	6.7	5.5	5.5	6.0	6.4	6.1	5.7	(14)
(15)	Precipitation	PrecAvg	16.0	0.9	0.7	0.7	0.9	1.3	2.1	2.1	2.2	1.7	1.4	1.1	0.9	(15)
(16)		PrecMax	20.5	1.9	2.2	1.5	1.7	3.7	5.2	3.9	3.4	3.9	2.9	2.2	1.6	(16)
(17)		PrecMin	9.1	0.2	0.2	0.1	0.0	0.3	0.2	0.6	0.7	0.6	0.2	0.2	0.4	(17)
(18)		PrecStd	2.8	0.4	0.4	0.4	0.5	0.7	1.1	1.0	0.7	0.9	0.7	0.5	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.6	36.4	47.3	64.0	82.5	87.9	88.9	86.9	75.7	61.9	43.7	36.1	(19)
(20)			MCWB	32.6	32.5	38.4	50.4	62.4	69.1	70.0	68.9	61.7	52.2	40.4	33.2	(20)
(21)		2%	DB	28.7	32.8	42.0	57.9	76.6	83.4	85.5	80.9	69.4	56.4	39.9	33.2	(21)
(22)			MCWB	27.1	30.2	34.9	47.0	59.2	66.9	68.5	66.8	58.6	49.0	37.7	31.3	(22)
(23)		5%	DB	24.6	28.8	38.5	54.2	71.1	79.4	82.6	75.8	64.2	51.9	36.5	30.2	(23)
(24)			MCWB	23.6	26.6	33.2	44.7	56.5	64.6	66.9	63.8	55.7	46.8	34.3	28.5	(24)
(25)		10%	DB	20.7	24.6	35.4	50.3	65.7	75.3	79.1	71.7	59.7	47.8	33.3	26.2	(25)
(26)			MCWB	19.5	22.5	31.3	42.0	53.9	62.6	65.6	61.9	53.6	44.2	31.4	24.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.5	33.5	38.9	52.7	64.3	71.1	72.5	70.7	63.1	53.1	41.1	33.9	(27)
(28)			MCD B	33.4	35.7	46.4	60.8	79.9	83.7	84.1	83.4	72.9	59.2	43.3	35.5	(28)
(29)		2%	WB	27.0	30.3	35.8	48.6	60.4	68.2	70.7	67.6	59.4	50.1	37.8	31.3	(29)
(30)			MCD B	28.4	32.8	40.5	56.7	73.7	80.6	81.9	76.9	67.1	55.0	39.7	33.2	(30)
(31)		5%	WB	23.5	26.7	33.7	45.3	57.5	66.1	69.1	65.5	56.7	46.9	34.5	28.6	(31)
(32)			MCD B	24.5	28.8	37.8	52.7	69.1	76.7	79.5	73.8	62.8	51.4	36.2	30.2	(32)
(33)		10%	WB	19.6	22.5	31.7	42.5	54.7	64.0	67.2	63.3	54.3	44.2	31.9	24.8	(33)
(34)			MCD B	20.8	24.5	35.1	49.1	64.7	73.2	76.5	70.2	59.2	47.7	33.2	26.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	13.5	16.2	16.8	16.8	18.4	17.0	18.1	16.3	13.8	10.7	11.2	12.4	(35)
(36)			MCDBR	13.1	13.1	16.5	22.0	26.8	23.3	23.9	22.8	20.2	16.6	9.8	12.1	(36)
(37)			MCWBR	12.7	11.6	11.7	13.8	14.1	10.4	9.9	10.1	10.4	10.7	8.5	11.4	(37)
(38)		5% WB	MCD BR	13.0	12.5	14.7	20.2	24.5	20.6	21.1	20.2	18.1	15.6	9.2	12.0	(38)
(39)			MCWBR	12.6	11.3	10.7	13.5	13.6	9.9	9.7	9.9	10.3	10.6	8.5	11.3	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.262	0.301	0.353	0.388	0.391	0.398	0.414	0.415	0.362	0.328	0.279	0.249	(40)
(41)		taud		1.999	2.031	1.981	2.085	2.295	2.342	2.296	2.259	2.378	2.308	2.093	1.946	(41)
(42)		Ebn at Noon		146	209	235	252	264	264	255	244	239	205	140	98	(42)
(43)		Edh at Noon		15	26	38	42	37	36	37	35	26	20	14	10	(43)
(44)	All-Sky Solar Radiation	RadAvg		141	428	914	1330	1596	1699	1679	1221	730	364	139	73	(44)
(45)		RadStd		10	29	52	143	130	139	148	115	78	46	26	7	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page